

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No Cummins® service tools required.

Additional Service Items

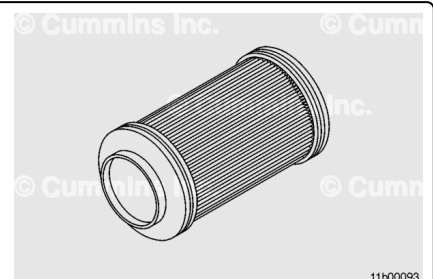
- 46 mm socket.

General Information

The aftertreatment diesel exhaust fluid (DEF) dosing unit filter is designed to prevent foreign objects, which may be suspended in the DEF, from entering the dosing system.

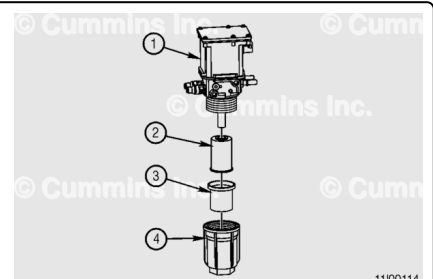
Debris can cause permanent damage and premature malfunction to either the aftertreatment DEF dosing unit or the aftertreatment DEF dosing valve. The aftertreatment DEF dosing unit filter is a maintenance item.

For handling incorrect or contaminated DEF, contact a Cummins® Authorized Repair Location.



The aftertreatment DEF dosing unit filter consists of the following components:

1. Aftertreatment DEF dosing unit
2. Aftertreatment DEF dosing unit element
3. Aftertreatment DEF dosing unit frost protection membrane
4. Aftertreatment DEF dosing unit filter housing.



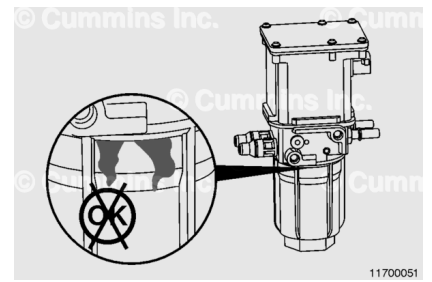
Initial Check

Locate the aftertreatment DEF dosing unit on the vehicle.

Inspect the plastic filter housing.

Inspect the area around the seal and filter housing for signs of leakage.

DEF leaks leave a white deposit. If deposits are found, see the Clean and Inspect for Reuse section of this procedure.



Preparatory Steps

⚠ WARNING ⚠

Diesel exhaust fluid (DEF) contains urea. Do not get the substance in your eyes. In case of contact, immediately flush eyes with large amounts of water for a minimum of 15 minutes. Do not swallow. In the event the DEF is ingested, contact a physician immediately. Reference the Materials Safety Data Sheet (MSDS) for additional information.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

- Allow for DEF to recirculate and depressurize after engine shut down and prior to DEF system servicing. This can take up to 15 minutes. There is always a chance of residual pressure being present. Open fittings slowly to allow any pressure to bleed off before removing any connections.
- Disconnect the batteries. Refer to the equipment manufacturer service information.

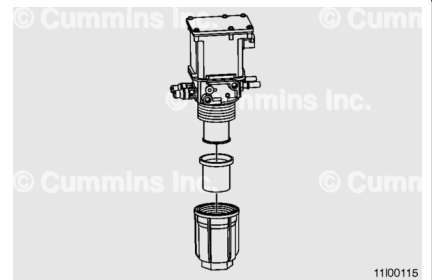
Note : Use compressed air to remove any loose debris.

- Use a clean, damp cloth to remove any contamination, to reduce the risk of contaminants entering the DEF dosing unit.

Remove

⚠ CAUTION ⚠

A small amount of dirt and debris can be very harmful to the DEF dosing system. Dirt or other foreign objects must not get into the DEF dosing unit.



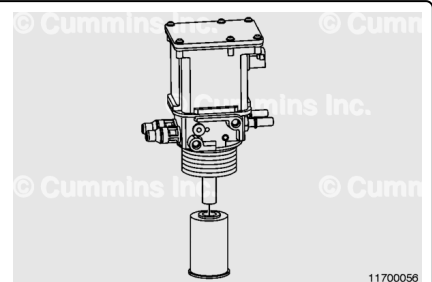
Note : There may be residual DEF in the filter housing. A collection container placed below the DEF filter housing is recommended.

Use a 46 mm socket to remove the DEF filter housing.

Remove the frost protection membrane and inspect. A small amount of DEF could leak out of it.

Remove the filter element from the unit by twisting, while pulling. Absorb the dripping DEF with a dry clean lint-free cloth and dry the unit.

Discard the frost protection membrane and filter element.



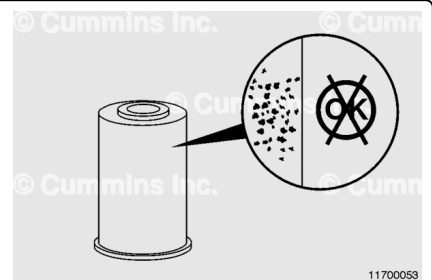
If there is the possibility that contaminated DEF has gone through the DEF dosing system, check the DEF filter prior to discarding it.

Check the dosing unit filter for evidence of contaminated DEF. Use visual and aroma characteristics of the filter to determine if contaminated fluid has passed through the dosing system.

Use the following procedure for further information on contaminated DEF. Refer to Procedure 011-056 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-056.html>)

Inspect the DEF filter for debris. If debris is evident, also check:

- DEF tank filter: See equipment manufacturer service information.
- The aftertreatment DEF dosing unit inlet connector: Refer to Procedure 011-058 in Section 11. (</qs3/pubsys2/xml/en/procedures/493/493-011-058.html>)

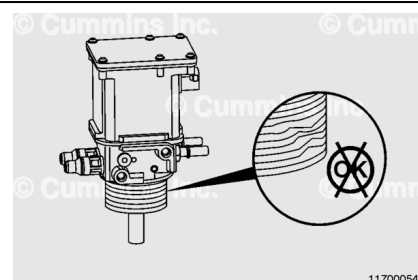


Clean and Inspect for Reuse

Inspect the aftertreatment DEF dosing unit threads. This is especially important if the aftertreatment DEF dosing unit cap was damaged.

If the aftertreatment DEF dosing unit threads are damaged, replace the entire aftertreatment DEF dosing unit.

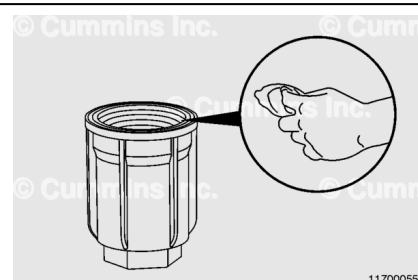
Verify the frost protection membrane groove is clean and free of debris. Clean using a clean, damp cloth, warm water, and a mild detergent.



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Note : Never operate the vehicle with the DEF filter or filter housing removed.

Clean the new aftertreatment DEF dosing unit filter housing and threads on the dosing unit with warm water and a clean cloth.



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Install

⚠ CAUTION ⚠

Use only DEF to lubricate the DEF filter o-ring. The use of any other lubricant can cause damage.

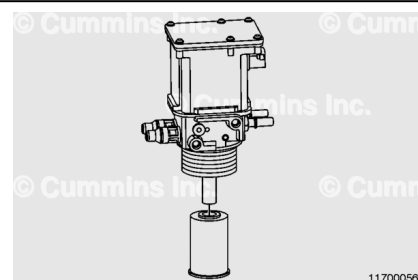
⚠ CAUTION ⚠

Confirm the filter is correctly seated. DEF system damage will occur if not seated.

Lubricate the new filter o-ring with DEF.

Slide the new filter element onto the DEF dosing unit.

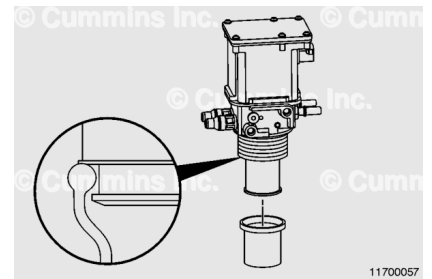
Push up on the filter until a click is heard.



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Install a new frost protection membrane over the filter element.

Note : The sealing bead of the frost protection membrane must sit completely in the groove of the pump housing.



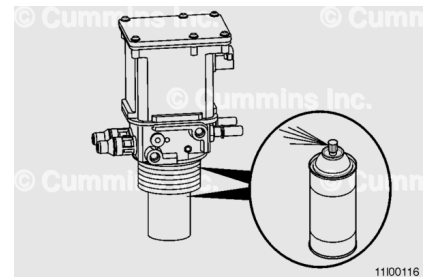
Note : The provided lubricant must be used when installing the filter housing.

Lubricate the top foam ring of the filter housing before torquing the filter onto the pump housing.

Spray the supplied lubricant to the threads on the pump housing and sealing bead of the frost protection membrane.

Install and tighten the filter housing with a 46 mm socket.

Torque Value: 80 N·m [59 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

Do not use the flow test portion of recommended Cummins® electronic service tool or equivalent Diesel Exhaust Fluid Doser Pump Override Test to check the system for leaks. This will spray DEF into the exhaust system at temperatures too low to evaporate, resulting in deposit formations in the exhaust system.

- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.

